EXPERIMENT 9

INPUT CODING

```
clc;
clear;
close all;

% Original signal
f = 40;                 % Signal frequency
t = 0:0.0001:0.1;
x = sin(2*pi*f*t);

% Sampling frequencies
Fs1 = 160;
Fs2 = 60;

% Sampling at 160 Hz
n1 = 0:1/Fs1:0.1;
x1 = sin(2*pi*f*n1);

% Sampling at 60 Hz
n2 = 0:1/Fs2:0.1;
x2 = sin(2*pi*f*n2);

% Calculate alias frequency
fa = abs(f - round(f/Fs2)*Fs2);

fprintf('Original Frequency = %d Hz\n', f);
fprintf('Sampling Frequency = %d Hz\n', Fs2);
fprintf('Predicted Alias Frequency = %.2f Hz\n', fa);

% Plot original signal and sampled signals
figure;

subplot(2,1,1);
plot(t,x,'b');
hold on;
stem(n1,x1,'r','filled');
title('Adequate Sampling (Fs = 160 Hz)');
xlabel('Time (seconds)');
ylabel('Amplitude');
legend('Original Signal','Sampled Signal');
grid on;

subplot(2,1,2);
plot(t,x,'b');
hold on;
stem(n2,x2,'r','filled');
title('Undersampling and Aliasing (Fs = 60 Hz)');
xlabel('Time (seconds)');
ylabel('Amplitude');
```

```
legend('Original Signal','Sampled Signal');
grid on;
```

OUTPUT

```
Original Frequency = 40 Hz
Sampling Frequency = 60 Hz
Predicted Alias Frequency = 20.0
```



